

1 Gain law

(bars denote normalized quantities: lengths by R , $\bar{a} = a/a_o$, $\bar{f} = a_o f$; $0 < \bar{a} < 1$; $e_\theta = \theta - \hat{\theta}$)

$$\bar{a}_{\text{true}}(\bar{w}) = 1 - \frac{9}{8\bar{w}} + \frac{1}{2\bar{w}^3} - \frac{57}{100\bar{w}^4} + \frac{1}{5\bar{w}^5} + \frac{7}{200\bar{w}^{11}} - \frac{1}{25\bar{w}^{12}}$$

$$\bar{a}'_{\text{true}}(\bar{w}) > 0 \quad \text{on} \quad \bar{w} > 1 \quad \implies \quad \bar{a}' = b_{\text{true}}(\bar{a})(1 - \bar{a})^2 \quad \text{exactly,} \quad b_{\text{true}} := \frac{\bar{a}'_{\text{true}}}{(1 - \bar{a}_{\text{true}})^2}$$

$$\bar{a}_{\text{true}}(1) = 0, \quad \bar{a}'_{\text{true}}(1) = 1 \quad \implies \quad b_{\text{true}}(\bar{a} = 0) = 1$$

$$1 - \bar{a}_{\text{true}} = \frac{9}{8\bar{w}} + \mathcal{O}(\bar{w}^{-3}) \quad \implies \quad b_{\text{true}}(\bar{a} \rightarrow 1) = \frac{8}{9}$$

The slope is evaluated at a HEIGHT, not at the gain state. Integrating the law once gives the level it is evaluated through:

$$\bar{a}(u) = 1 - \frac{1}{b u}$$

$$\bar{a}'(u) = b(1 - \bar{a}(u))^2 = \frac{1}{b u^2}$$

$$u[k] = \bar{w}_d[k] - \bar{w}_0 - \delta \hat{w}_3[k] \quad (\text{act: the estimated ACTUAL height})$$

$$u[k] = \bar{w}_d[k] - \bar{w}_0 \quad (\text{cmd: the COMMANDED height})$$

$$a_w = a_o \bar{a}_w, \quad a_o = \frac{\Delta t}{\gamma_N R}$$

2 Jacobians

$$\frac{\partial \bar{a}'}{\partial \bar{a}_w} = 0$$

$$\frac{\partial \bar{a}'}{\partial b} = -\frac{1}{b^2 u^2} = -\frac{\bar{a}'}{b}$$

$$\frac{\partial \bar{a}'}{\partial \delta \bar{w}_3} = +\frac{2}{b u^3} = +\frac{2\bar{a}'}{u} \quad (\text{act; 0 for cmd})$$

3 True dynamics

$$(\delta \bar{w}_j[k] := \delta \bar{w}[k - d + j - 1], \quad j = 1, \dots, d + 1, \quad d = 2)$$

$$\bar{w}[k] = \bar{w}_d[k] - \delta \bar{w}_3[k]$$

$$\bar{w}_d[k + 1] = \bar{w}_d[k] + \Delta \bar{w}_d[k]$$

$$\delta \bar{w}_1[k + 1] = \delta \bar{w}_2[k]$$

$$\delta \bar{w}_2[k + 1] = \delta \bar{w}_3[k]$$

$$\delta \bar{w}_3[k + 1] = \lambda_c \delta \bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k]$$

$$\bar{a}_w[k + 1] = \bar{a}_w[k] + \bar{a}'_{\text{true}}[k] \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right]$$

$$b[k + 1] = b[k]$$

$$F_{dw}[k] = \bar{f}_{dw}[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{f}_{dw}[k - i]$$

$$\varepsilon_w[k] = \bar{a}_w[k] \bar{f}_T[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{a}_w[k - i] \bar{f}_T[k - i] + (1 - \lambda_c) n_w[k]$$

4 Process model

$$\mathbf{x} = [\delta \bar{w}_1, \delta \bar{w}_2, \delta \bar{w}_3, \bar{a}_w, b]^\top \quad (d = 2)$$

5 Estimator

$$\delta \hat{w}_1[k + 1] = \delta \hat{w}_2[k]$$

$$\delta \hat{w}_2[k + 1] = \delta \hat{w}_3[k]$$

$$\delta \hat{w}_3[k + 1] = \lambda_c \delta \hat{w}_3[k]$$

$$\hat{a}_w[k + 1] = \hat{a}_w[k] + \frac{1}{\hat{b}[k] \hat{u}[k]^2} \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \hat{w}_3[k] \right]$$

$$\hat{b}[k + 1] = \hat{b}[k]$$

$$\hat{u}[k] = \bar{w}_d[k] - \bar{w}_0 - \delta \hat{w}_3[k]$$

6 Error dynamics

(True – Estimator; coefficients at the estimate)

$$\begin{aligned}
e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k] \\
e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\
e_{\delta\bar{w}_3}[k+1] &= \lambda_c e_{\delta\bar{w}_3}[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
e_{\bar{a}_w}[k+1] &= \left[1 + \frac{1}{\hat{b}\hat{u}^2} F_{dw} \right] e_{\bar{a}_w}[k] \\
&\quad + \left[\frac{1-\lambda_c}{\hat{b}\hat{u}^2} + \frac{2}{\hat{b}\hat{u}^3} (\Delta\bar{w}_d + (1-\lambda_c)\delta\hat{w}_3) \right] e_{\delta\bar{w}_3}[k] \\
&\quad - \frac{1}{\hat{b}^2\hat{u}^2} (\Delta\bar{w}_d + (1-\lambda_c)\delta\hat{w}_3) e_b[k] \\
&\quad + \frac{1}{\hat{b}\hat{u}^2} \varepsilon_w[k] \\
e_b[k+1] &= e_b[k]
\end{aligned}$$

$$F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} & 0 \\ 0 & 0 & \frac{1-\lambda_c}{\hat{b}\hat{u}^2} & 1 + \frac{1}{\hat{b}\hat{u}^2} F_{dw} & -\frac{1}{\hat{b}^2\hat{u}^2} \\ 0 & 0 & +\frac{2}{\hat{b}\hat{u}^3} (\Delta\bar{w}_d + (1-\lambda_c)\delta\hat{w}_3) & \times (\Delta\bar{w}_d + (1-\lambda_c)\delta\hat{w}_3) & 1 \end{bmatrix}$$

$$\mathbf{q}[k] = \left[0, 0, -\varepsilon_w, \frac{1}{\hat{b}\hat{u}^2} \varepsilon_w, 0 \right]^\top$$

7 Measurement and innovation

$$\begin{aligned}
\nabla_d \bar{w}_d[k] &= \bar{w}_d[k] - \bar{w}_d[k-d] \\
y_1[k] &= \delta\bar{w}_1[k] + n_w[k] \\
y_2[k] &= \bar{a}_w[k-d] + n_a[k]
\end{aligned}$$

$$\bar{a}_w[k-d] = \bar{a}_w[k] - \frac{1}{b u^2} \nabla_d \bar{w}_d[k]$$

$$\begin{aligned}
\frac{\partial y_2}{\partial \bar{a}_w} &= 1 \\
\frac{\partial y_2}{\partial \delta\bar{w}_3} &= -\frac{2}{\hat{b}\hat{u}^3} \nabla_d \bar{w}_d \quad (\text{act}; 0 \text{ for cmd}) \\
\frac{\partial y_2}{\partial b} &= +\frac{1}{\hat{b}^2\hat{u}^2} \nabla_d \bar{w}_d
\end{aligned}$$

$$H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2}{\hat{b}\hat{u}^3}\nabla_d\bar{w}_d & 1 & +\frac{1}{\hat{b}^2\hat{u}^2}\nabla_d\bar{w}_d \end{bmatrix}$$

$$L[k] = P^f H^\top (H P^f H^\top + R)^{-1}$$

$$e_{y_1}[k] = e_{\delta\bar{w}_1}[k] + n_w[k]$$

$$e_{y_2}[k] = e_{\bar{a}_w}[k] - \frac{2}{\hat{b}\hat{u}^3}\nabla_d\bar{w}_d e_{\delta\bar{w}_3}[k] + \frac{1}{\hat{b}^2\hat{u}^2}\nabla_d\bar{w}_d e_b[k] + n_a[k]$$

8 Process and measurement noise

$$q_3 = -\bar{a}_w \bar{f}_T$$

$$q_4 = -\frac{1}{\hat{b}\hat{u}^2} q_3$$

$$\gamma(\bar{w}) = \gamma_N c(\bar{w})$$

$$\sigma_{\bar{f}_T}^2 = a_o^2 \frac{4k_B T \gamma(\bar{w})}{\Delta t}$$

$$Q_{33} = \frac{4k_B T a_o}{R} \bar{a}_w$$

$$Q_{34} = Q_{43} = -\frac{1}{\hat{b}\hat{u}^2} Q_{33}$$

$$Q_{44} = \frac{1}{\hat{b}^2\hat{u}^4} Q_{33}$$

$$Q_{55} = 0$$

$$R = \text{diag}(R_1, R_2)$$

$$R_1 = \sigma_{n_w}^2$$

$$R_2 = K_{\text{var}} I F_{\text{var}} (\bar{a}_w + \xi)^2 + d Q_{44}$$

$$K_{\text{var}} = \frac{2a_{\text{cov}}}{2 - a_{\text{cov}}}$$

$$\sigma_{\delta\bar{w}_r}^2 = C_{dpmr} \frac{4k_B T a_o}{R} (\bar{a}_w + \xi)$$

$$\xi = \frac{C_n \sigma_{n_w}^2}{C_{dpmr} (4k_B T a_o / R)}$$

Seed and prior

$$\begin{aligned}\hat{b}[0] &= 1 \\ P_{55}[0] &= P_{bb} = \left[\sup_{\bar{w} \in \mathcal{E}} |b_{\text{true}}(\bar{w}) - 1| \right]^2 \\ \hat{a}_w[0] &= 1 - \frac{1}{\hat{b}[0] \hat{u}[0]}\end{aligned}$$

$$\begin{aligned}\left. \frac{\partial \bar{a}}{\partial b} \right|_u &= \frac{1}{b^2 u} = \frac{1 - \bar{a}}{b} \\ P_{44}[0] &= \left(\frac{1 - \hat{a}_w[0]}{\hat{b}[0]} \right)^2 P_{bb} + \text{floor}_a^2 \\ P_{45}[0] &= \frac{1 - \hat{a}_w[0]}{\hat{b}[0]} P_{bb}\end{aligned}$$

Observability

Gate 1. Both routes for b :

$$\begin{aligned}\text{state route} \quad \frac{\partial \bar{a}_w[k+1]}{\partial b} &= -\frac{1}{\hat{b}^2 \hat{u}^2} (\Delta \bar{w}_d + (1 - \lambda_c) \delta \hat{w}_3) \\ \text{measurement route} \quad \frac{\partial y_2}{\partial b} &= +\frac{1}{\hat{b}^2 \hat{u}^2} \nabla_d \bar{w}_d\end{aligned}$$

Gate 2. Time signatures on the gain channel:

$$\begin{aligned}e_b &\longrightarrow \text{scaled by the commanded DISPLACEMENT} \\ e_{\delta \bar{w}_3} &\longrightarrow \text{scaled by the commanded displacement AND by } 1/\hat{u} \\ e_{\bar{a}_w} &\longrightarrow \text{scaled by the applied force } F_{dw}\end{aligned}$$

Gate 3. Reduction on $(\bar{a}_w, b)$, with $h := 1$ and $\Sigma := \Delta \bar{w}_d + (1 - \lambda_c) \delta \hat{w}_3$:

$$\begin{aligned}A_2 &= \begin{bmatrix} 1 + \frac{F_{dw}}{\hat{b} \hat{u}^2} & -\frac{\Sigma}{\hat{b}^2 \hat{u}^2} \\ 0 & 1 \end{bmatrix}, \quad C_2 = \begin{bmatrix} 1 & \frac{\nabla_d \bar{w}_d}{\hat{b}^2 \hat{u}^2} \end{bmatrix} \\ \det O_2 &= -\frac{\Sigma}{\hat{b}^2 \hat{u}^2} - \frac{\nabla_d \bar{w}_d}{\hat{b}^2 \hat{u}^2} \frac{F_{dw}}{\hat{b} \hat{u}^2} \\ \det O_2 = 0 &\iff \Sigma = -\frac{\nabla_d \bar{w}_d F_{dw}}{\hat{b} \hat{u}^2}\end{aligned}$$

In a hold both Σ and $\nabla_d \bar{w}_d$ vanish and the condition is met identically: b carries no information there, in this writing as in the state writing.